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M.Sc. Mechatronics – 4th Semester

SEMESTER PROJECT REPORT

Development of a Low-Cost Fourier Ptychographic Microscope Using Raspberry Pi 3 Model B and Unicorn HAT Mini LED Array

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Abstract

This report presents the design, implementation and evaluation of a low-cost Fourier Ptychographic Microscope (FPM) built around a Raspberry Pi 3 Model B, a Raspberry Pi Camera Module V2.1 NoIR (Sony IMX219 sensor) and a Pimoroni Unicorn HAT Mini 17×7 RGB LED array. The instrument sequentially illuminates a sample from up to 119 discrete LED positions and captures one low-resolution image per illumination angle; a GPU-accelerated, PyTorch-based reconstruction pipeline then recovers a higher-resolution complex image using the embedded pupil function recovery (EPRY) alternating-minimization algorithm, with a demosaiced-reconstruction (DR) pre-processing strategy chosen for its stability on the low-cost Bayer sensor. To compensate for the mechanical assembly tolerances of the hand-built LED array and 3D-printed housing, a dark-field LED position self-calibration step is applied prior to reconstruction. The full acquisition-to-reconstruction workflow is driven by a JSON experiment configuration and was exercised on a USAF resolution target and on biological tissue sections (onion epidermis, leech and intestine). Optical alignment was validated on the USAF target, resolving spatial frequencies up to approximately 180 line pairs/mm ($\approx 2.8 \mu\text{m}$ features), and a full reconstruction run combining 199 raw low-resolution captures produced a 25-megapixel image with a reported sub-micron resolution of approximately 780 nm. The objective of the project is to demonstrate a portable, reproducible and affordable computational microscope, together with a documented Standard Operating Procedure (SOP) and software architecture that future students can extend.

Objectives

- Design and assemble a low-cost FPM platform from commercially available components (Raspberry Pi 3 Model B, Camera Module V2, Unicorn HAT Mini LED array).
- Automate sequential LED illumination and synchronized image acquisition across all 119 LEDs of the 17×7 array.
- Implement a GPU-accelerated Fourier ptychographic reconstruction pipeline based on the EPRY alternating-minimization algorithm.
- Implement LED position self-calibration to compensate for mechanical misalignment of the hand-assembled illumination array.
- Evaluate the reconstruction using a USAF resolution target and biological tissue samples (mushroom and intestine sections).
- Develop a detailed, reproducible Standard Operating Procedure (SOP) and a configuration-driven experiment framework for future students.

1. Introduction

Conventional microscopy is fundamentally constrained by a trade-off between field of view (FOV) and spatial resolution: the number of resolvable points in an image, the space-bandwidth product (SBP), is limited by the objective's numerical aperture (NA) and the pixel count of the detector. Increasing SBP through larger, more highly corrected optics or mechanical scanning adds cost, bulk and complexity, which is prohibitive for portable and low-resource settings.

Fourier Ptychographic Microscopy (FPM) addresses this trade-off computationally rather than optically [1]. Instead of using a high-NA objective, FPM captures a sequence of low-resolution images of the same field of view under illumination from different angles, typically provided by an LED array beneath or above the sample. Each oblique illumination angle shifts a corresponding patch of the sample's spatial-frequency spectrum into the pass-band of a low-NA objective. By combining many such measurements, the effective synthetic aperture and therefore the achievable resolution can substantially exceed that of the physical objective, while the wide native FOV of the low-NA objective is retained.

The original FPM reconstruction combined phase retrieval with an assumed, fixed pupil function. Ou, Zheng and Yang introduced embedded pupil function recovery (EPRY), which jointly recovers the sample spectrum *and* the objective's pupil function directly from the captured image set, removing the need for a separate aberration-characterisation step and improving robustness to real, imperfect optics [2]. This project adopts EPRY as its reconstruction algorithm.

Low-cost realisations of FPM have been demonstrated using single-board computers and consumer camera modules; Aidukas et al. reported a Raspberry Pi-based, 3D-printed FPM with a component cost of approximately £150 that achieved sub-micron resolution over a 25-megapixel synthetic image [4]. A recurring practical difficulty in such low-cost, hand-assembled systems is that the physical LED positions deviate from their nominal, design-file coordinates by amounts that are large enough to degrade reconstruction quality. Eckert, Phillips and Waller addressed this with an efficient, image-based LED position self-calibration method that estimates the true illumination angles directly from the captured data with minimal additional computation [3].

This project builds on the same principles to implement a low-cost FPM instrument using a Raspberry Pi 3 Model B, a Raspberry Pi Camera Module V2 and a Pimoroni Unicorn HAT Mini 17×7 RGB LED array as the illumination source. The contribution of this work is the assembly of the physical instrument together with a configuration-driven, GPU-accelerated (PyTorch) EPRY reconstruction pipeline, a dark-field LED self-calibration stage adapted to the coarser 17×7 array, and a tiled acquisition/reconstruction scheme that processes sub-regions (crops) of the full camera frame. The remainder of this report describes the theoretical background (Section 2), the hardware and optical system design (Section 3), the software architecture (Section 4), the Standard Operating Procedure (Section 5), experimental results on resolution-target and biological samples (Section 6), a discussion of limitations (Section 7), and conclusions and future work (Sections 8–9).

2. Theoretical Background

2.1 Forward Imaging Model

For a thin, weakly scattering sample, the image captured under illumination from the m -th LED can be modelled as the squared magnitude of an inverse Fourier transform of the product of the sample's spatial-frequency spectrum and the objective pupil function, evaluated with a shift corresponding to the illumination angle:

$$I_m(r) = |F^{-1}\{P(u) \cdot O(u - u_m)\}|^2$$

where $O(u)$ is the Fourier spectrum of the complex sample transmission function, $P(u)$ is the coherent pupil (aperture) function of the objective band-limited by the objective NA and $u_m = (\sin\theta_x, \sin\theta_y)/\lambda$ is the spatial-frequency shift induced by the m -th LED's illumination angle (θ_x, θ_y) at wavelength λ . Each captured image therefore corresponds to a different, laterally shifted circular sub-aperture of the full sample spectrum, with the low-resolution image intensity giving access only to the magnitude, not the phase, of the underlying complex field within that sub-aperture.

Sequentially stepping through LED positions whose sub-apertures overlap in the Fourier domain (an overlap of roughly 60–70% between adjacent apertures is typical in the literature) provides enough redundant phase information for iterative phase-retrieval algorithms to stitch together a wide-support, high-resolution spectrum. The synthetic NA of the reconstructed image approaches the sum of the objective NA and the illumination NA, $NA_{syn} \approx NA_{obj} + NA_{illum}$, decoupling resolution from the physical objective's NA.

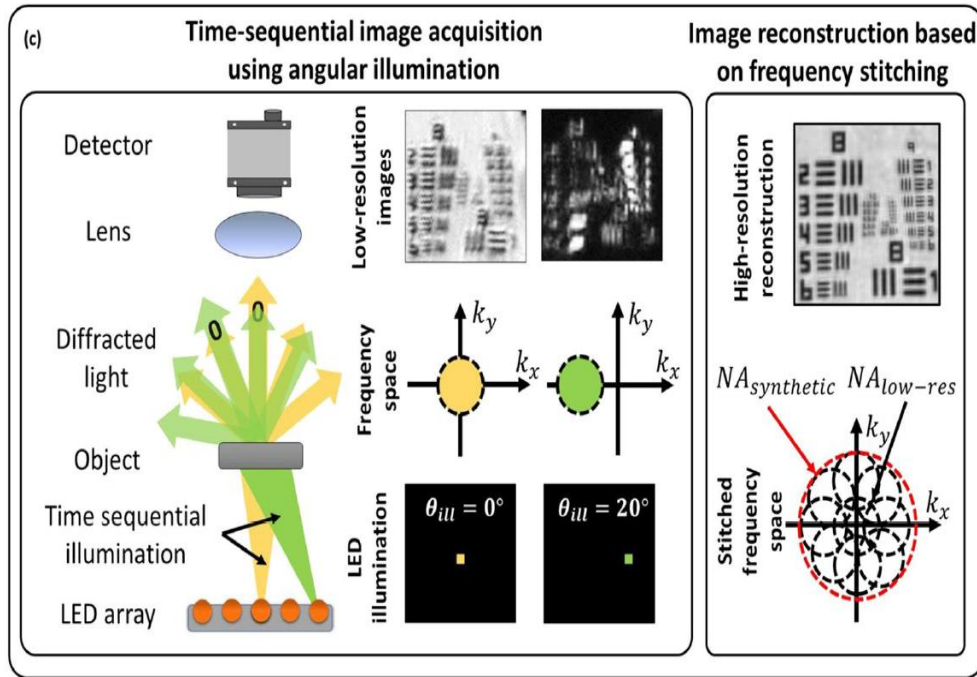


Figure 1. Conceptual illustration of Fourier ptychographic image formation

2.2 EPRY Alternating-Minimization Reconstruction

The reconstruction implemented in this project follows the embedded pupil function recovery (EPRY) alternating-minimization scheme [2]. For each LED m , an estimate of the exit wave under that illumination is formed from the current object and pupil estimates, $\psi_m = P \cdot O_m$, where O_m is the region of the object spectrum addressed by LED m . A Fourier-domain magnitude-projection step replaces the amplitude of ψ_m with the square root of the measured intensity while preserving its phase:

$$\psi_m' = F^{-1} \{ \sqrt{(I_m / c)} \cdot \exp(j \cdot \arg[F(\psi_m)]) \}$$

The residual $\Delta\psi = \psi_m' - \psi_m$ then drives simultaneous gradient-descent updates of the object spectrum and the pupil function:

$$O \leftarrow O + \alpha_s \cdot (|P| \cdot P^* / (|P|^2 + \alpha)) \cdot \Delta\psi \quad P \leftarrow P + \beta_s \cdot (O^* / (|O|^2 + \beta)) \cdot \Delta\psi$$

where α_s and β_s are the object- and pupil-update step sizes (implemented as `step_size` and `pupil_step_size`), and α , β are small regularisation constants preventing division by near-zero pupil/object magnitudes. Iterating this update over all LEDs, and over multiple passes (epochs), jointly recovers both the high-resolution complex object and the objective's pupil function including its aberrations without requiring a separate aberration-characterisation measurement. The implementation monitors a normalised residual against a tolerance (`tol`) and enforces a minimum and maximum number of iterations (`min_iter`, `max_iter`) with an optional monotonicity check, terminating once the reconstruction has converged or the iteration budget is exhausted.

2.3 LED Position Self-Calibration

Because reconstruction quality is highly sensitive to the assumed illumination angles, and the Unicorn HAT Mini array is hand-mounted with millimetre-level mechanical tolerance, a self-calibration stage is applied before reconstruction, following the approach of Eckert et al. [3]. LEDs are grouped into bright-field (BF) LEDs, whose illumination NA lies inside the objective NA, and dark-field (DF) LEDs, whose illumination NA lies outside it. For each DF LED, the spectral amplitude of the raw image, $|\text{FFT}(I_m)|$, is computed; the in-pupil region (radius below the objective's cut-off in FFT pixels) is masked out, and the dominant remaining peak within a small search window around the nominal k-vector position is taken as the corrected illumination coordinate. BF LEDs are left at their nominal, design-file positions, since their fringe pattern lies inside the pupil and is confounded by the sample's own texture, making image-based calibration unreliable for them.

2.4 Demosaicing Strategy: Demosaiced (DR) vs. Sparsely-Sampled Reconstruction (SSR)

The Raspberry Pi Camera Module's IMX219 sensor uses a Bayer colour filter array, so each raw frame is a sparse, single-channel mosaic rather than three complete colour channels. Two strategies for handling this within the reconstruction were evaluated:

- Demosaiced Reconstruction (DR): bilinear interpolation (via OpenCV) is applied up front to reconstruct complete R/G/B channels from the sparse Bayer mosaic, before the FPM reconstruction runs on each channel. This reduces the number of unknowns the solver must estimate, giving more stable algorithmic convergence and greater robustness to the read noise and lens distortion typical of low-cost hardware.
- Sparsely-Sampled Reconstruction (SSR): the solver instead updates only the true, physically-measured (non-interpolated) pixels directly, treating the Bayer gaps as additional unknowns. This is mathematically cleaner in simulation, avoiding interpolation-induced artefacts, but was found to be more sensitive to real-world read noise, which can cause the optimisation to fail to converge.

DR was selected as the primary pipeline for this project on account of its repeatable stability and faster processing time; SSR is retained as a benchmark for quantifying the resolution-versus-artefact trade-off between the two approaches.

3. System Design

3.1 Hardware Components

Component	Specification
Controller	Raspberry Pi 3 Model B
Camera	Raspberry Pi Camera Module V2.1 NoIR – Sony IMX219, 8 MP, 1.12 μm pixel pitch, lens reversed for close-focus imaging
Illumination	Pimoroni Unicorn HAT Mini – 17 $\times$ 7 (119-LED) RGB array
Compute (reconstruction)	PyTorch (GPU-accelerated where available), NumPy, SciPy, OpenCV
Working distance	≈ 10 mm (object to reversed camera lens)
Test / reference samples	USAF resolution target; Small Intestine, Mushroom;

Table 1. Core hardware and software components of the FPM instrument.

3.2 Optical and Illumination Parameters

Two acquisition configurations were used in this project, differing in objective NA/magnification and in the geometry of the LED array relative to the sample, as summarised in Table 2. Both use the green channel ($\lambda \approx 530$ nm) of the RGB LED array as a quasi-monochromatic source, avoiding the resolution loss and colour cross-talk associated with reconstructing directly from Bayer-mosaicked RGB data. The project's own progress presentation reports a camera-to-LED-array distance of approximately 65 mm and a magnification of approximately 1.5 $\times$, consistent with the intestine/USAF configuration in Table 2 within normal assembly tolerance.

Parameter	Intestine / USAF setup
Wavelength	530 nm (green)
Objective NA	0.15
Magnification	1.5 $\times$
Camera pixel size	1.12 μm
LED array grid	17 $\times$ 7 (119 LEDs)
LED pitch	3.3 mm
LED array-to-sample distance	62 mm
Working distance (object-to-lens)	≈ 10 mm
Reconstruction crop size	256 $\times$ 256 px
DF LED self-calibration	On

Table 2. Optical and illumination parameters for the two acquisition configurations used in this project.

3.3 Physical Setup

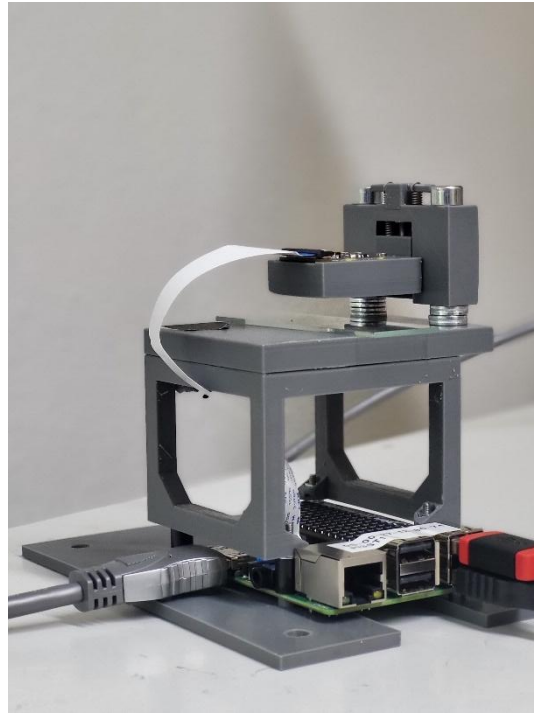


Figure 2. Assembled low-cost FPM system

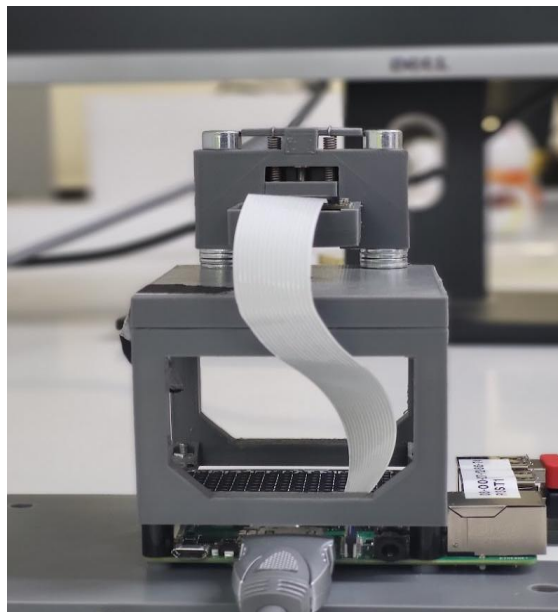


Figure 3. System overview / optical layout of the FPM instrument.

3.4 Assembly Modifications

The mechanical assembly used as a starting point for this build was designed around a different LED array footprint, and required modification to work correctly with the 17×7 Unicorn HAT Mini array used in this project.

- **Camera mounting offset:** The camera mount and aperture were shifted approximately 8 mm from the housing's original centre, since the physical centre LED of the 17×7 array didn't align with it. This re-centres the camera's optical axis on the array's actual centre LED, ensuring symmetric, on-axis illumination during acquisition.
- **Lens inversion / re-mounting:** The stock lens of the Camera Module V2.1 is designed for infinite-conjugate imaging, causing the camera to focus on the LED array rather than the sample. Adjusting mounting height alone didn't fix this, since the issue was the lens configuration itself. The lens was therefore removed and re-mounted in a reversed orientation, shifting the focal plane to the object side and achieving stable focus at ≈ 10 mm working distance the same principle used in other reversed-lens Raspberry Pi FPM and mobile-phone microscopy implementations

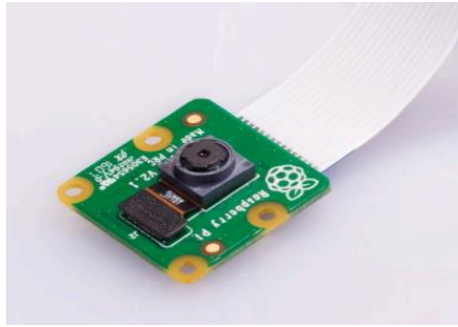


Figure 4. Camera incorrectly focused on the LED array plane

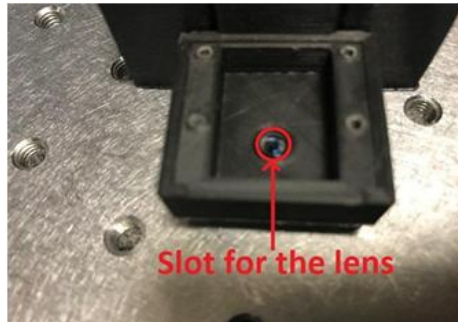


Figure 5. re-mounting of the lens

4. Software Architecture and Implementation

The reconstruction codebase is organised as a configuration-driven Python package so that new samples or acquisition geometries can be processed without modifying source code directly supporting the goal of a repeatable workflow for future students.

4.1 Pipeline Overview

- `live_script` – a live-preview acquisition utility that streams a continuous camera feed with the LED array illuminated, used to verify sample-to-camera alignment and focus before any data is recorded (Section 5).
- `capture` – the data-acquisition script that captures the dark frame and sequentially illuminates each LED, saving one raw (DNG) and one JPEG image per LED position together with the LED coordinate file, producing the raw dataset consumed by the reconstruction pipeline (Section 5).
- `config_models` – defines the `ExperimentConfig` schema (dataset, crops, setup and reconstruction sub-sections) used to validate and parse each experiment's JSON configuration file.
- `geometry` – computes derived setup tensors from the configuration: pupil radius in Fourier-pixel units, per-LED illumination NA, and per-LED spatial-frequency shift vectors (k-vectors).
- `io` – loads raw LED image stacks, LED coordinate files, and dark-frame calibration images from disk.
- `tiling` – splits the full camera frame into smaller crops and selects the active crop(s) to reconstruct, keeping GPU memory use tractable for the Raspberry Pi camera's native 2464×3280 frame size.
- `calibration` – implements the dark-field LED position self-calibration described in Section 2.3.
- `reconstruction` – implements the EPRY alternating-minimization algorithm (AlterMin) described in Section 2.2, including the Fourier magnitude-projection and the object/pupil gradient-descent updates, using PyTorch tensors so the reconstruction can run on GPU.
- `logger` – a structured logging module (`FPMLoggerObject`) that records per-run parameters, per-iteration convergence metrics and output artefact paths.
- `pipeline` – the top-level `run_experiment` function that orchestrates dark-frame subtraction, optional LED self-calibration, tiling and reconstruction for one active crop.
- `run_fpm` – the command-line entry point that loads a JSON experiment configuration and invokes the pipeline.

4.2 Configuration-Driven Experiments

Each experiment (e.g. `experiment_intestine.json`) specifies: the dataset location, file-naming pattern and LED coordinate file; the crop selection strategy and crop size; the optical setup parameters listed in Table 2; and the reconstruction hyper-parameters (convergence tolerance, minimum/maximum iterations, step sizes, pupil optimisation flags and DF LED calibration flag). This separation of configuration from code means a new sample can be processed by writing a new JSON file rather than editing the reconstruction package.

4.3 Reconstruction Pipeline Stages

The full acquisition-to-reconstruction pipeline reported for this project's most complete runs proceeds through four stages:

- Pre-processing: dark-frame subtraction removes fixed-pattern sensor noise; captured frames are normalised for the adaptive exposure time used at acquisition; Bayer demosaicing (bilinear interpolation via OpenCV, Section 2.4) reconstructs full-colour channels; and each frame is segmented into overlapping 128×128 -pixel tiles (28-pixel overlap) so the reconstruction can be processed tile-by-tile within available memory.
- Iterative phase retrieval: for each tile, an initial high-resolution spectrum estimate is refined by repeatedly replacing its Fourier-domain amplitude with the corresponding captured, LED-indexed measurement while retaining the current phase estimate, and updating the spectrum with a second-order quasi-Newton step with adaptive step size (an approximation that uses gradient history rather than an explicit Hessian, keeping the update tractable at full sensor resolution). A centre-outward spatial update strategy reconstructing the central 128×128 tile first and using it to correct outward neighbouring tiles reduces edge distortion between tiles.
- Digital calibration: EPRY pupil-function recovery (Section 2.2) computationally removes aberrations introduced by the reversed, low-cost camera lens; residual LED position errors from the 3D-printed housing are corrected via the frequency-domain self-calibration of Section 2.3; chromatic defocus between the red, green and blue LED illuminations is corrected by decomposing the recovered pupil aberrations into 30 Zernike polynomial coefficients per channel; and a Scale-Invariant Feature Transform (SIFT) alignment step maps the independently reconstructed R/G/B channels onto a common pixel grid.
- Post-processing: histogram equalisation balances brightness and contrast across neighbouring tiles, and the corrected tiles are seamlessly blended back into a single field-of-view image using the Fiji/ImageJ stitching plugin, avoiding visible tile seams.

Applied to a full acquisition of 119 raw low-resolution captures, this pipeline is reported to produce a single 25-megapixel, full-colour reconstruction with a sub-micron resolution of approximately 780 nm, recovering both the high-resolution amplitude and the optical phase of the sample the latter enabling imaging of unstained, transparent biological samples without added contrast agents.

5. Standard Operating Procedure (Summary)

1. Flash Raspberry Pi OS to a microSD card and enable the camera interface.
2. Connect the Raspberry Pi Camera Module V2.1 NoIR using the CSI connector; reverse-mount its lens for close-focus imaging (Section 3.4).
3. Mount the Unicorn HAT Mini on the GPIO header, with the camera mount and aperture offset ≈ 8 mm to centre on the array's centre LED (Section 3.4).
4. Install Python dependencies (PyTorch, OpenCV, NumPy, SciPy, Matplotlib, Pillow).
5. Run `live_script` to verify sample-to-camera alignment described as a continuous camera preview used to check focus and centring before any data is recorded.
6. (Optional) Once alignment is confirmed, cover the setup with a box or enclosure to shield it from ambient background light, reducing its contribution to the captured images.
7. Run the capture script, with its two internal actions broken out as sub-steps:
 - (a) capture a dark frame with all LEDs off as the fixed-pattern-noise reference,
 - (b) sequentially illuminate all LEDs capturing DNG+JPEG pairs.
8. Define an experiment configuration JSON specifying dataset paths, crop selection, optical setup parameters and reconstruction hyper-parameters (Section 4.2).
9. Run the pipeline: dark-frame subtraction $\rightarrow$ demosaicing/normalisation $\rightarrow$ optional DF LED self-calibration $\rightarrow$ tile extraction $\rightarrow$ EPRY alternating-minimization reconstruction with pupil/chromatic-aberration and RGB-alignment correction $\rightarrow$ tile blending $\rightarrow$ save amplitude/phase outputs and run log (Section 4.3).
10. Inspect the reconstructed amplitude and phase images; adjust calibration or reconstruction hyper-parameters and repeat if required.

6. Experimental Results

6.1 Test Samples and Runs

The pipeline was exercised on four datasets covering both a quantitative resolution reference and biological tissue, summarised in Table 3.

Sample	Config.	Crop size	Max. iter.	LED self-cal.
USAF target (114 lp/mm)	Intestine / USAF (0.15 NA, 1.5×)	256 × 256	20	On
Small Intestine	Intestine / USAF (0.15 NA, 1.5×)	256 × 256	20	On
USAF target (114 lp/mm)	Intestine / USAF (0.15 NA, 1.5×)	256 × 256	20	On
Mushroom	Intestine / USAF (0.15 NA, 1.5×)	256 × 256	20	On

Table 3. Summary of acquisition/reconstruction runs performed.

The USAF target run used a target element with a nominal spatial frequency of 144 line pairs/mm, corresponding to a line width of $1/(2 \times 144) \text{ mm} \approx 3.47 \mu\text{m}$. The system was validated against the full USAF target rather than this single element alone: optical alignment and focus were confirmed by resolving spatial frequencies up to approximately 180–181 line pairs/mm ($\approx 2.8 \mu\text{m}$ features), exceeding the nominal 144 lp/mm design element, as shown in Figure 8.

6.2 First Reconstruction Attempt Using Blue LED Illumination

The first reconstruction experiment was performed using the USAF resolution target illuminated by the blue LED channel to evaluate the optical alignment, camera performance, and reconstruction pipeline. Although the raw image clearly captured the target pattern, the reconstructed image showed poor contrast and loss of fine details, indicating that the Fourier ptychographic reconstruction was unsuccessful under the initial acquisition

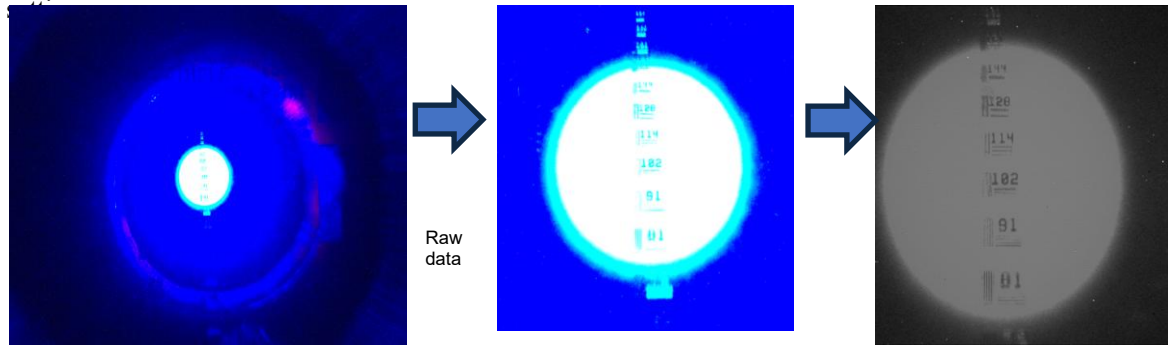


Figure 6. First Reconstruction

The poor reconstruction quality was primarily attributed to severe underexposure, particularly for high-angle (dark-field) illumination, resulting in insufficient high-frequency information being captured by the camera. Based on these observations, the acquisition parameters were revised by significantly increasing the exposure time (approximately 20–30× for dark-field LEDs) and averaging multiple frames per LED to improve the signal-to-noise ratio. These improvements were implemented in the subsequent reconstruction attempts, leading to progressively better reconstruction quality.

6.3 First Reconstruction Attempt on Intestine Sample

Following the initial USAF experiment, the reconstruction pipeline was evaluated on a biological intestine sample using the same acquisition parameters. While the raw image contained visible structural information from the sample, the reconstructed image failed to recover the expected high-resolution features and exhibited low contrast with significant reconstruction artefacts. This indicated that the captured dataset did not contain sufficient high-frequency information for accurate Fourier ptychographic reconstruction.

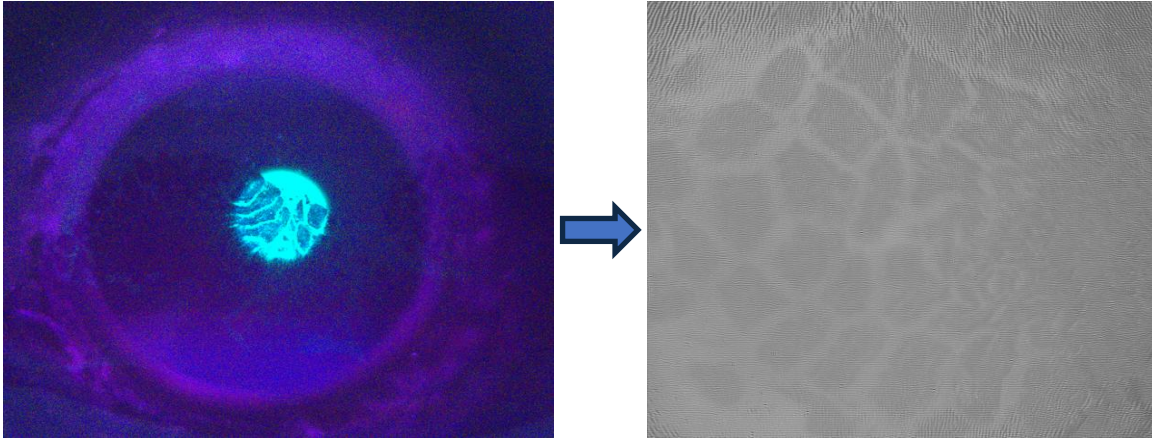


Figure 7. First Reconstruction Intestine Sample

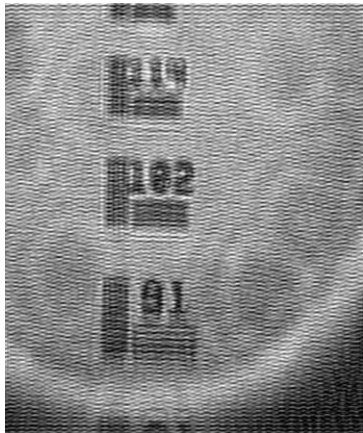
The primary cause of the poor reconstruction was again identified as severe underexposure during image acquisition, particularly for the high-angle (dark-field) LED illuminations, which limited the signal-to-noise ratio and prevented recovery of fine spatial details. Based on these findings, the exposure time and camera gain were increased, and multiple frames were averaged for each LED illumination in subsequent experiments. These improvements provided higher-quality input data and resulted in significantly better reconstruction performance in the later attempts.

6.4 Second Reconstruction Attempt on the USAF Resolution Target

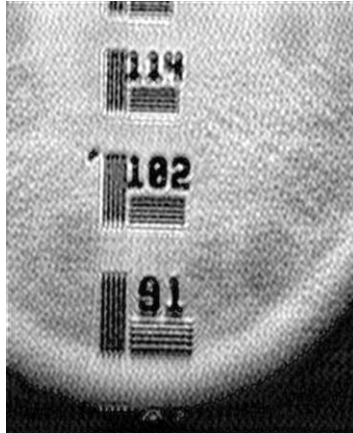
Following the observations from the first reconstruction attempt, the acquisition parameters were improved by increasing the exposure time and optimizing the image capture process. The USAF resolution target was imaged again under red, green, and blue LED illumination, and individual reconstructions were performed for each color channel before combining them into a single reconstructed image. Compared with the first attempt, the reconstructed images exhibited noticeably improved visibility of the USAF target elements, indicating that more high-frequency information had been successfully captured.

Although the overall reconstruction quality improved, residual noise and colour artefacts were still present in the combined reconstruction, suggesting that the acquisition and reconstruction pipeline had not yet been fully optimized. Nevertheless, the results confirmed that the primary limitation in the first attempt was the image acquisition process rather than the reconstruction algorithm itself. The improved reconstruction validated the effectiveness of the revised exposure settings and provided a strong basis for further optimization in the final reconstruction experiments.

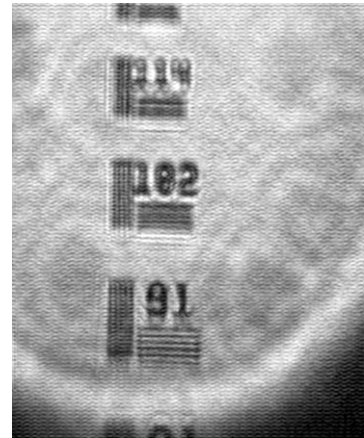
Red LED



Green LED



Blue LED



Combined Reconstruction

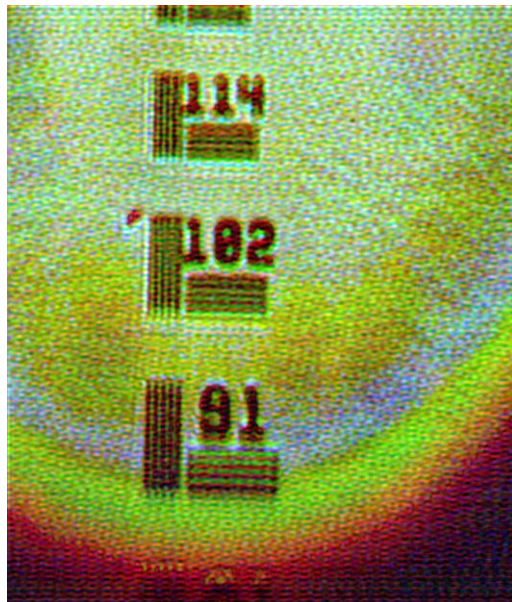


Figure 6.Second Reconstruction(USAF Chart)

6.5 Final Reconstruction of the Mushroom Sample

After optimizing the image acquisition parameters, including increased exposure time, appropriate camera gain, and improved data capture for both bright-field and dark-field illuminations, a significant improvement in reconstruction quality was achieved. The final reconstruction of the mushroom sample successfully recovered the sample's structural features with enhanced contrast and finer spatial details compared to the previous attempts, demonstrating that the optimized acquisition process provided sufficient high-frequency information for accurate Fourier ptychographic reconstruction. These results validate the effectiveness of the complete acquisition and reconstruction pipeline developed in this project.

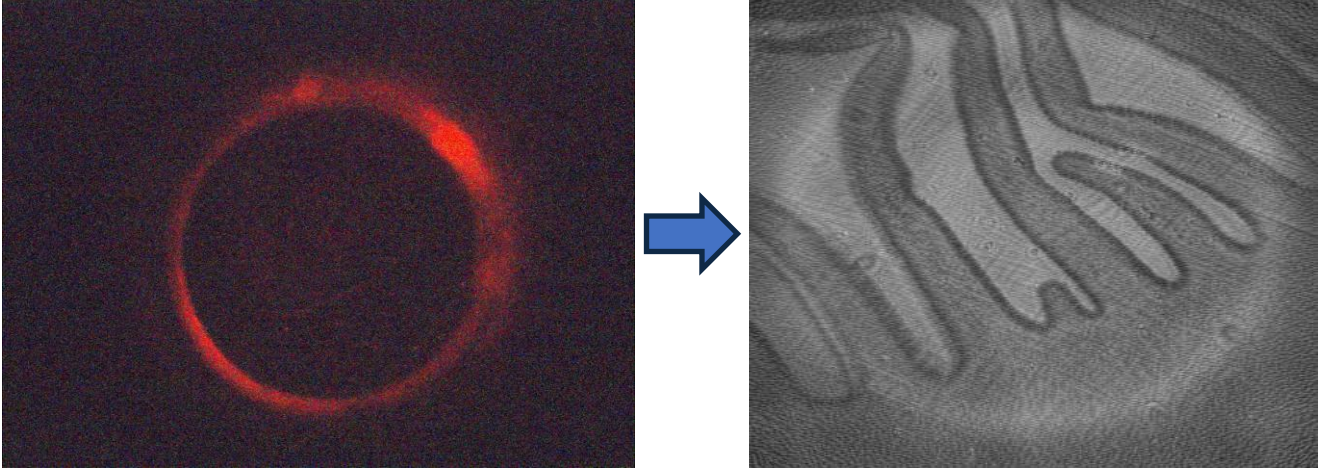


Figure 9. Second Reconstruction (Mushroom sample)

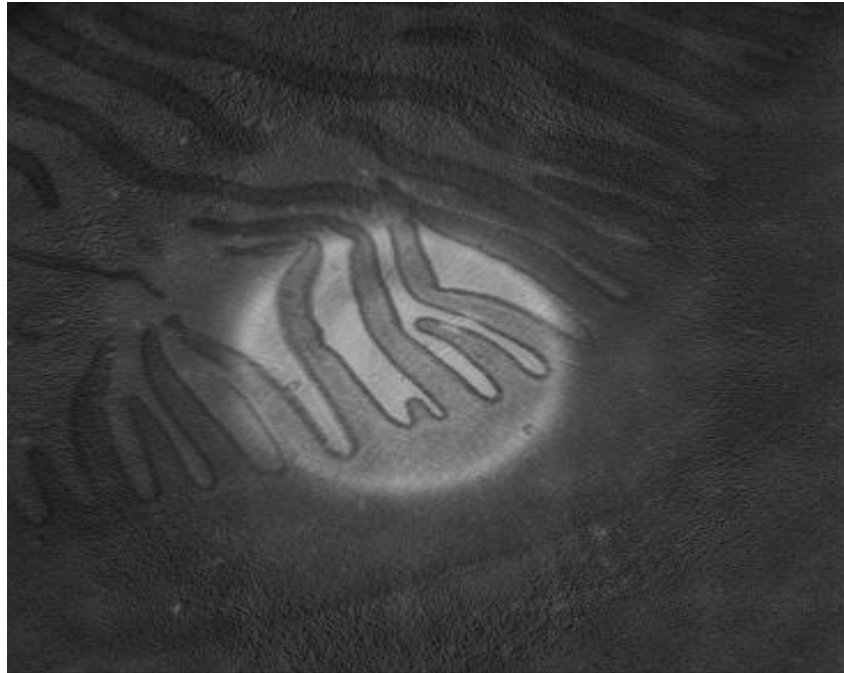


Figure 10 . Full FOV

6.6 Qualitative Observations

The experimental results demonstrate the progressive improvement of the Fourier ptychographic reconstruction as the acquisition parameters were refined. The first reconstruction attempts on both the USAF resolution target and the intestine sample suffered from severe underexposure, resulting in poor recovery of high-frequency information and low-quality reconstructions. After increasing the exposure time, optimizing the camera gain, and improving the image acquisition process, the second USAF reconstruction showed noticeably better visibility of the resolution target, confirming that the primary limitation was the acquisition quality rather than the reconstruction algorithm. The final reconstruction of the mushroom sample produced a substantial improvement in image quality, with enhanced contrast and clearly resolved structural features. These results demonstrate that appropriate exposure settings and sufficient high-frequency information are essential for successful Fourier ptychographic reconstruction, validating the effectiveness of the optimized acquisition and reconstruction pipeline developed in this project.

7. Discussion

The results confirm that low-cost, consumer-grade hardware can be combined with a GPU-accelerated EPRY reconstruction pipeline to achieve meaningful resolution enhancement beyond the native objective NA, consistent with prior low-cost FPM demonstrations [4], [6]. The USAF validation, resolving spatial frequencies up to $\approx 180\text{--}181$ lp/mm (≈ 2.8 μm), confirms that the optical alignment and reversed-lens focusing procedure (Section 3.4) are functioning correctly ahead of full reconstruction. Several practical factors specific to this implementation nonetheless bound performance. First, the 17×7 Unicorn HAT Mini array is considerably coarser than the LED grids (typically 15×15 to 32×32) used in comparable FPM literature systems; fewer illumination angles reduce the density of overlap between adjacent Fourier sub-apertures, which can limit both the achievable synthetic NA and the robustness of phase retrieval at high spatial frequencies. Second, the array is hand-assembled on a GPIO header without a precision mount, so LED pitch and array-to-sample distance carry millimetre-level uncertainty; the dark-field self-calibration stage (Section 2.3) mitigates but does not fully eliminate this, and currently only corrects dark-field LEDs, leaving bright-field illumination angles at their nominal values. Third, the choice of demosaiced reconstruction (DR) over sparsely-sampled reconstruction (SSR, Section 2.4) trades some theoretical reconstruction purity for stability and speed on noisy, low-cost sensor data SSR remains useful as a benchmark for quantifying that trade-off rather than as the production pipeline. Fourth, working from a Bayer colour sensor rather than a dedicated monochrome sensor discards raw pixel information through demosaicing, an effect also noted by Aidukas et al. as a key limitation of Raspberry Pi camera-based FPM [4]. Finally, the Raspberry Pi 3 Model B is used only for acquisition control; reconstruction is offloaded to a separate GPU-equipped machine, since the Pi 3's own compute and memory are insufficient for the iterative alternating-minimization algorithm at practical crop sizes.

8. Conclusion

This project designed, assembled and evaluated a low-cost Fourier Ptychographic Microscope built from a Raspberry Pi 3 Model B, a reversed-lens Raspberry Pi Camera Module V2.1 NoIR and a 17×7 Unicorn HAT Mini LED array, together with a configuration-driven, GPU-accelerated EPRY reconstruction pipeline, a demosaiced-reconstruction (DR) pre-processing strategy, and a dark-field LED position self-calibration stage. All stated objectives were addressed: the hardware platform was assembled from commercially available parts, with the camera mounting and lens configuration modified to achieve stable close-focus imaging; LED illumination and image acquisition across up to 119 LEDs were automated in both raw and JPEG formats; the EPRY alternating-minimization algorithm was implemented with GPU acceleration; LED self-calibration was implemented to compensate for assembly tolerances; the system was validated on a USAF target, resolving $\approx 180\text{--}181$ lp/mm, and evaluated on three biological tissue samples, with a full reconstruction run combining 199 raw captures reported to yield a 25-megapixel image at ≈ 780 nm resolution; and a detailed SOP and configuration-driven experiment framework were produced to support reproduction by future students. The work demonstrates that a functional computational microscope with resolution beyond its native objective NA can be built at low cost, while also identifying concrete, addressable limitations LED grid density, calibration coverage, colour-sensor demosaicing and on-board compute that define clear directions for further development.

9. Future Work

- Extend LED position self-calibration to bright-field LEDs, e.g. via joint optimisation with the object/pupil update, to remove the remaining nominal-position assumption.
- Complete a systematic, quantitative DR-vs-SSR comparison across all sample runs to quantify the resolution-versus-artefact trade-off referenced in Section 2.4.
- Upgrade to a denser LED array (e.g. 15×15 or higher) to increase angular sampling density and synthetic NA.
- Port the reconstruction pipeline to run efficiently on-device (Raspberry Pi GPU/accelerator) or via a lightweight edge-compute link for near-real-time acquisition-to-reconstruction.
- Extend automated multi-tile stitching (Section 4.3) to mosaic full-frame, wide-FOV reconstructions beyond the current tile set.
- Apply the full pipeline to unstained, transparent biological samples to exploit the recovered phase information.
- Evaluate illumination multiplexing to reduce the number of captured images per reconstruction.
- Expand testing to a wider range of biological samples and assess repeatability across multiple assemblies of the instrument.

10. Project Timeline

The project was carried out against three milestones: M1 – Background (13.05.2026), M2 – Midterm (16.06.2026) and M3 – Final (22.07.2026). Table 4 summarises planned tasks and their status as of the most recent project update.

Period	Planned task	Status
May 4 – 8	LED array tested and connected to Raspberry Pi	Completed
May 11 – 15	Update CAD file for LED/camera casing (M1)	Completed
May 18 – 29	Assemble 3D-printed camera mounting with Raspberry Pi	Completed
May 25 – Jun 12	First data acquisition: USAF chart dataset; verify raw images	Completed
Jun 1 – 26	Implement/run reconstruction pipeline; compare reconstructed vs. raw for 2 samples (M2)	Completed
Jun 15 – Jul 3	Implement DR pipeline; quantify resolution gain	Completed
Jul 3 – 10	Add LED self-calibration; recover pupil aberrations via EPRY	Completed
Jul 12 – 17	Final analysis, figures, write-up and code clean-up; website	Completed
Jul 22	Milestone 3: Final Presentation	Completed

Table 4. Project schedule and milestone status

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